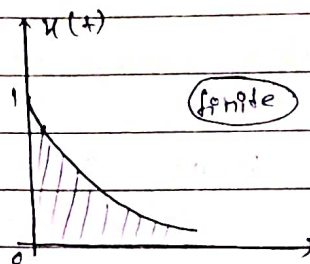


UNIT - IV* Laplace Transform -

WKT, $x(t) \xrightarrow{F.T} X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt < \infty$

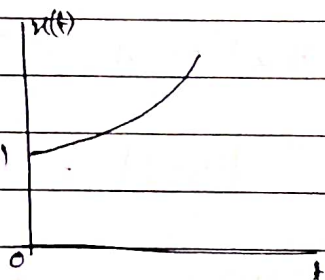
If $x(t) = e^{-2t} u(t) \Rightarrow$

$\hookrightarrow$ FT is finite
(or $< \infty$)



But $x(t) = e^{4t} u(t) \Rightarrow$

$\hookrightarrow$ FT is not finite
(or $\neq \infty$)



So, multiply by $e^{-\sigma t}$, where $\sigma > 4$, i.e., e^{-5t}

$\therefore e^{4t} \times e^{-5t} = e^{-t} < \infty$

Therefore, $F.T [x(t) e^{-\sigma t}] = \int_{-\infty}^{\infty} x(t) e^{-\sigma t} e^{-j\omega t} dt$

$$= \int_{-\infty}^{\infty} x(t) e^{-(\sigma + j\omega)t} dt$$

or, $\boxed{L.T [x(t)] = \int_{-\infty}^{\infty} x(t) e^{-pt} dt}$, where $p = \sigma + j\omega$

→ Based on the nature of the signal $x(t)$, there are the set of σ values only for which the Laplace Transform will exist, and this range of σ values is defined as the region of convergence (ROC)

→ ROC of Laplace transform is always denoted in terms of σ or $\text{Re}\{s\}$

* Condition for Laplace Transform —

$$\left| \int_{-\infty}^{\infty} x(t) e^{-\sigma t} e^{-j\omega t} dt \right| < \infty$$

$$\int_{-\infty}^{\infty} |x(t) e^{-\sigma t}| |e^{-j\omega t}| dt < \infty$$

$$\int_{-\infty}^{\infty} |x(t) e^{-\sigma t}| dt < \infty$$

→ existence condition

When $x(t) = \delta(t)$

$$\Rightarrow \int_{-\infty}^{\infty} |\delta(t) e^{-\sigma t}| dt$$

$$\Rightarrow \int_{-\infty}^{\infty} |\delta(t)| dt = 1, \text{ for all values of } \sigma$$

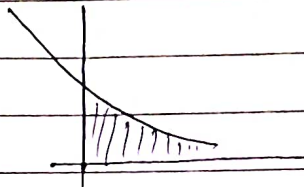
$$\therefore \delta(t) \xrightarrow{\text{L.T}} 1; \text{ ROC: } \forall \sigma$$

When $x(t) = u(t)$

$$\Rightarrow \int_{-\infty}^{\infty} |u(t) e^{-\sigma t}| dt$$

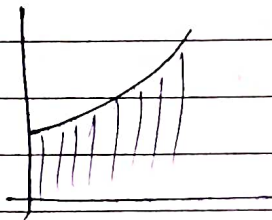
$$\Rightarrow \int_0^{\infty} |e^{-\sigma t}| dt$$

when $\sigma > 0$:



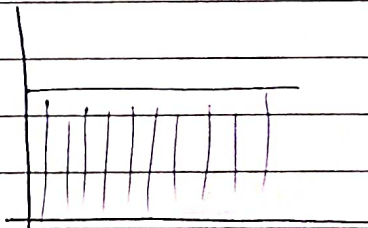
$< \infty$

when $\sigma < 0$:



$> \infty$

when $\sigma = 0$:



$> \infty$

So, $X(s) = \int_{-\infty}^{\infty} u(t) e^{-st} dt$

$\swarrow$
L.T $[u(t)]$

$$= \int_0^{\infty} e^{-st} dt$$

$$= \frac{1}{s}$$

; ROC: $\sigma > 0$ or $\text{Re}\{s\} > 0$

$j\omega$

s -plane

When $x(t) = u(-t)$

$$\Rightarrow X(s) = -\frac{1}{s} \quad ; \quad \text{ROC: } \sigma < 0$$

When $x(t) = -u(-t)$

$$\Rightarrow X(s) = \frac{1}{s} \quad ; \quad \text{ROC: } \sigma < 0$$

* Two L.T of different signals may be same but ROC is different.

* Properties of ROC —

- ① ROC of L.T can't include any poles but it is always bounded by poles.
- ② ROC of L.T is generally in the form of rectangular stripes
- ③ ROC of right-sided signal is always right sided. Vice-versa for left-sided signal.
- ④ If a signal $x(t)$ is to be Fourier transformable, the ROC of its L.T must be including $j\omega$ -axis or $\sigma = 0$ line, and if ROC includes $j\omega$ -axis, F.T can be calculated from the L.T expression by replacing $s = j\omega$.

$$e^{-at} u(t) \xrightarrow{\text{L.T}} \frac{1}{s+a}$$

$$\xrightarrow{\text{F.T}} \frac{1}{j\omega+a}$$

⑤ If the LTI system with impulse response $h(t)$ is to be stable, we must have L.T with its ROC including $j\omega$ -axis.

⑥ Combination of any number of right-sided signals will have a L.T which is the sum of all the individual L.T whereas ROC will be right sided bounded by greatest among all the poles.

⑦ Combination of any number of left-sided signals will have a L.T which is the combination of sum of the individual L.T with a ROC which is left sided bounded by least among all the poles.

⑧ For a finite duration signals having finite amplitude, ROC is in general entire s -plane.

⑨ For a two/both sided signal L.T may or may not be define, it is define if the the greatest pole of L.T of the right sided combination of signals is less than the least poles of the L.T of the left sided combination of the signals.

* Properties of Laplace Transform -

① Time Shifting :-

$$\begin{aligned} x(t) &\xrightarrow{L.T} X(s) \\ x(t - t_0) &\xrightarrow{L.T} X(s) e^{-st_0} \\ x(t + t_0) &\xrightarrow{L.T} X(s) e^{st_0} \end{aligned}$$

② Linearity :-

$$\begin{aligned} u_1(t) &\xrightarrow{L.T} X_1(s) \\ u_2(t) &\xrightarrow{L.T} X_2(s) \end{aligned}$$

e.g. $\Rightarrow e^{-j\omega_0 t} u(t) \xrightarrow{L.T} \frac{1}{s + j\omega_0} ; \text{ROC} : \sigma > 0$

$e^{j\omega_0 t} u(t) \xrightarrow{L.T} \frac{1}{s - j\omega_0} ; \text{ROC} : \sigma > 0$

$$\Rightarrow \underbrace{\frac{1}{2} e^{-j\omega_0 t} u(t) + \frac{1}{2} e^{j\omega_0 t} u(t)}_{\cos \omega_0 t u(t)} \xrightarrow{L.T} \underbrace{\frac{1}{2} \times \frac{1}{s + j\omega_0}}_{\sigma > 0} + \underbrace{\frac{1}{2} \times \frac{1}{s - j\omega_0}}_{\sigma > 0}$$

$\Rightarrow \cos \omega_0 t u(t) \xrightarrow{L.T} \frac{s}{s^2 + \omega_0^2} ; \text{ROC} : \sigma > 0$

③ Multiplication by 't' / Frequency shifting differentiation :-

$$\begin{aligned} x(t) &\xrightarrow{L.T} X(s) \\ t x(t) &\xrightarrow{L.T} -\frac{d}{ds} X(s) \end{aligned}$$

e.g. $\Rightarrow t u(t) \xrightarrow{L.T} \frac{1}{s^2}$

$$t^n u(t) \xrightarrow{L.T} \frac{n!}{s^{n+1}}$$

④ Time Differentiation property :-

$$x(t) \xrightarrow{L.T} X(s)$$

$$\frac{dx(t)}{dt} \xrightarrow{L.T} s X(s)$$

$$\frac{d^n x(t)}{dt^n} \xrightarrow{L.T} s^n X(s)$$

⑤ Time Integration Property :-

$$\int_{-\infty}^t x(\tau) d\tau \xrightarrow{L.T} \frac{1}{s} X(s)$$

* Shifting in s-Domain —

$$x(t) \xrightarrow{L.T} X(s) ; \text{ROC} : R(\text{rational})$$

$$e^{s_0 t} x(t) \xrightarrow{L.T} X(s - s_0) ; \text{ROC} : R + \text{Re}\{s_0\}$$

① Time Scaling :-

$$x(t) \xrightarrow{L.T} X(s) ; \text{ROC} : R$$

$$x(at) \xrightarrow{L.T} \frac{1}{|a|} X\left(\frac{s}{a}\right) ; \text{ROC} : aR$$

② Time Reversal :-

$$\begin{aligned} x(t) &\xrightarrow{L.T} X(s) && ; \text{ROC} : R \\ x(-t) &\xrightarrow{L.T} X(-s) && ; \text{ROC} : -R \end{aligned}$$

③ Inverse L.T :-

$$X(s) = \int_0^{\infty} x(t) e^{-st} dt \quad \rightarrow \text{Unilateral L.T}$$

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt \quad \rightarrow \text{Bilateral L.T}$$

$$\Rightarrow x(t) = \frac{1}{2\pi j} \int_{\sigma-j\omega}^{\sigma+j\omega} X(s) e^{st} ds \quad \rightarrow (\text{I.L.T})$$

④ Partial - fraction Expansion :-

$x(s)$: Rational function

$$x(s) = \frac{K(s-z_1)(s-z_2)\dots(s-z_m)}{(s-p_1)(s-p_2)\dots(s-p_n)}$$

when $\rightarrow m < n$: Proper Rational function

$$x(s) = \frac{A}{(s-p_1)} + \frac{B}{(s-p_2)} + \dots$$

when $\rightarrow m \geq n$: Improper Rational function

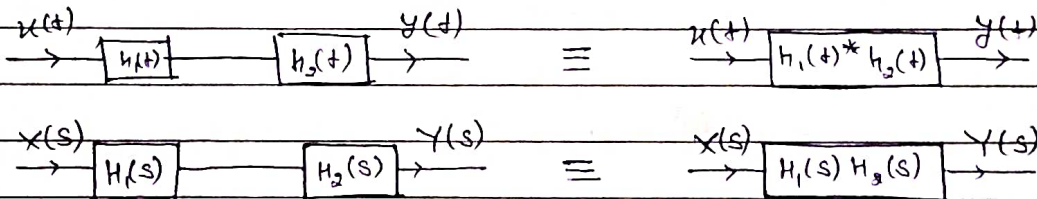
$$X(s) = \frac{N(s)}{D(s)} = Q(s) + \frac{R(s)}{D(s)}$$

$Q(s) \rightarrow$ Quotient polynomial in 's' with degree $(m-n)$

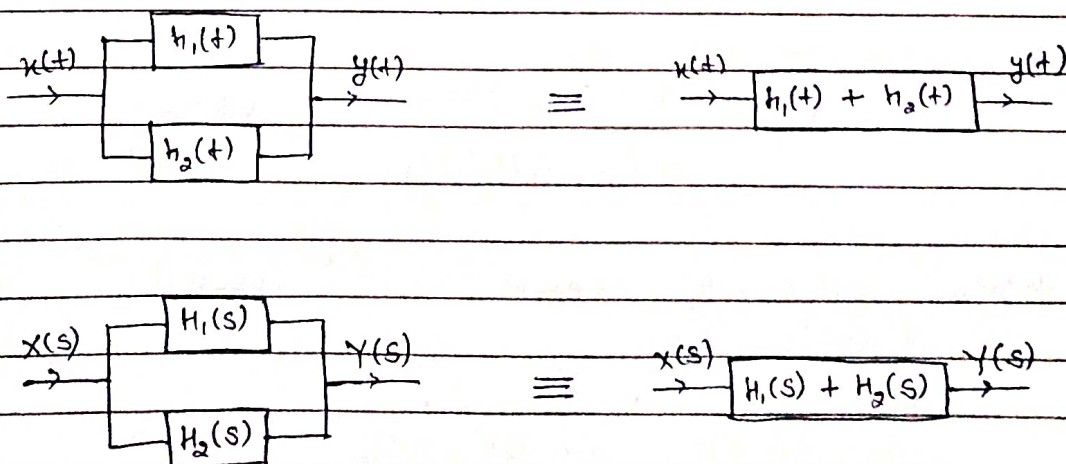
$R(s) \rightarrow$ Remainder polynomial in 's' with degree less than 'n'

$$* \frac{d^k s(t)}{dt^k} \xrightarrow{L.T} s^k ; k=1, 2, 3, \dots$$

* Cascade or Series Connection —



* Parallel Connection —

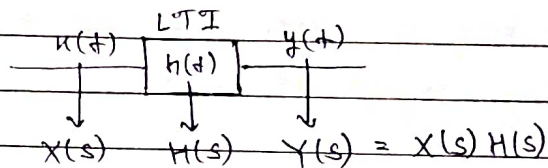


* Convolution Property -

$$x(t) \xrightarrow{L.T} X(s)$$

$$h(t) \xrightarrow{L.T} H(s)$$

$$\Rightarrow x(t) * h(t) \xrightarrow{L.T} X(s) H(s)$$

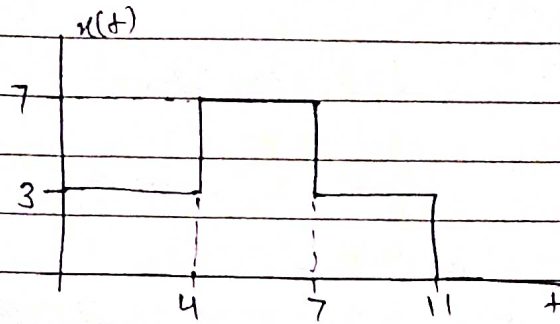


$$H(s) = \frac{Y(s)}{X(s)}$$

System function
or

System Transfer function

Q28 $\Rightarrow$ Find the Laplace transform -



Q29 $\Rightarrow$ Find Laplace transform of $x(t) = e^{-2t}u(t) + e^{-3t}u(t)$. Plot pole, zero and region of convergence.

Q30 $\Rightarrow$ Find L.T, plot pole-zero and ROC, for -

$$x(t) = e^{-a|t|}, \text{ for } a > 0 \text{ \& } a < 0$$

* Time-Differentiation Property -

$$x(t) \xrightarrow{\text{L.T}} X(s)$$

$$\frac{d}{dt} x(t) \xrightarrow{\text{L.T}} sX(s) - x(0^-)$$

$$\frac{d^2}{dt^2} x(t) \xrightarrow{\text{L.T}} s^2 X(s) - sx(0^-) - x'(0^-)$$

* Time-Integration Property -

$$\int_0^t x(\tau) d\tau \xrightarrow{\text{L.T}} \frac{1}{s} X(s)$$

$$\int_{-\infty}^t x(\tau) d\tau \xrightarrow{\text{L.T}} \frac{1}{s} X(s) + \frac{1}{s} \int_{-\infty}^0 x(\tau) d\tau$$

* Initial - Value Theorem —

$$x(t) \xrightarrow{L.T.} X(s)$$

$$x(0) = \lim_{s \rightarrow \infty} s X(s)$$

* Final - Value Theorem —

$$x(t) \xrightarrow{L.T.} X(s)$$

$$x(\infty) = \lim_{s \rightarrow 0} s X(s)$$

→ Initial value theorem can be applied only if the degree of numerator of laplace transform is less than the degree of the denominator of L.T, i.e., proper rational number. (It should be proper rational no.)

→ Final value theorem can be applied only if all the poles of $sX(s)$ lie in the left half of s -plane.

Q31 $\Rightarrow$ Find the inverse laplace transformation -

$$X(s) = \frac{2s + 4}{s^2 + 4s + 3}, \quad \text{Re}\{s\} > -1$$

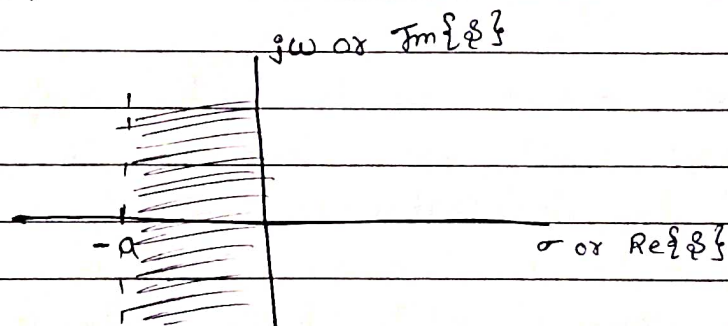
Q32 $\Rightarrow$ Find inverse laplace transformation -

$$X(s) = \frac{5s + 13}{s(s^2 + 4s + 13)}, \quad \text{Re}\{s\} > 0$$

Q33 $\Rightarrow$ Find inverse laplace transformation -

$$X(s) = \frac{s^2 + 2s + 5}{(s+3)(s+5)^2}, \quad \text{Re}\{s\} > -3$$

* stable system -



If system includes $j\omega$ -axis then, system is stable system.

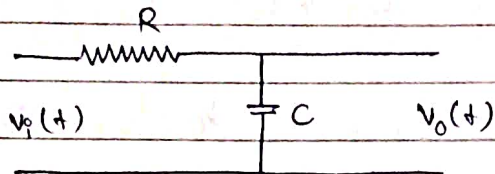
i.e., all poles in left plane $\rightarrow$ are stable system.

$\rightarrow$ when ROC includes $j\omega$ -axis, we directly write F.T from L.T by replacing 's' by 'jω'

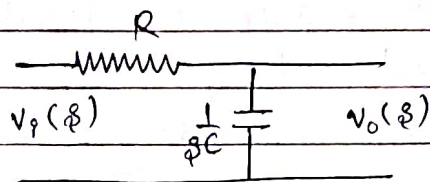
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Delta

(Unit -4) (Questions)

Q34 $\Rightarrow$ Find the system function and impulse Response.A $\Rightarrow$ System / transfer function $\Rightarrow h(t) = \frac{y(t)}{x(t)}$

Laplace Transform of Circuit :-



$$V_o(s) = \frac{1/sC}{R + 1/sC} V_i(s)$$

$$V_o(s) = \frac{1}{1 + sRC} V_i(s)$$

$$\text{So, } H(s) = \frac{V_o(s)}{V_i(s)} = \frac{1}{1 + sRC}$$

$$\text{Now, } H(s) \xrightarrow{\text{ILT}} h(t)$$

$$\frac{1}{sRC + 1} \xrightarrow{\text{ILT}} \frac{1}{RC} e^{-t/RC} u(t)$$

$$\therefore h(t) = \frac{1}{RC} e^{-t/RC} u(t)$$

Q35 $\Rightarrow y''(t) + y'(t) - 2y(t) = x(t)$ where,

$y(t)$: output & $x(t)$: input

Find system function.

A $\Rightarrow$ Taking Laplace transform -

$$[\cancel{s^2 Y(s)} - \cancel{s Y(0)} - \cancel{Y'(0)}] + [\cancel{s Y(s)} - \cancel{Y(0)}] - 2Y(s) = X(s)$$

As initial conditions are not given i.e., zero -

$$s^2 Y(s) + s Y(s) - 2Y(s) = X(s)$$

$$\frac{Y(s)}{X(s)} = \frac{1}{s^2 + s - 2} = \frac{1}{(s+2)(s-1)}$$

Taking inverse Laplace -

$$h(t) = \frac{y(t)}{x(t)} = \frac{1}{3} (e^t u(t) + e^{-2t} u(t))$$

[check again]

(H.W)

Q36 $\Rightarrow Y'(t) + ay(t) = x(t)$ with $y(0) = y_0$ and

$$x(t) = k e^{-bt} u(t)$$

Find the response of the system.

A $\Rightarrow y(t) = y_0 e^{-at} + \frac{k}{a-b} (e^{-bt} - e^{-at}) u(t) \rightarrow \text{(final answer)}$